

Warm-up 1

Compute the following matrix-vector multiplications:

$$\begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} =$$

$$\begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} =$$

During class 1

Find the eigenvalues and eigenvectors of the following matrix.

$$\begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix}$$